

LAREDO, TX, USA

WMO: 722520

Lat: 27.533N

Lon: 99.467W

Elev: 494

StdP: 14.44

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12907

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.0	39.0	14.0	11.4	53.2	18.6	14.3	54.1	26.6	65.7	24.3	64.4	8.6	0	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.8	105.1	74.7	102.9	74.7	101.2	74.7	79.5	92.0	78.7	91.1	78.2	90.4	12.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	143.5	82.8	75.9	137.8	82.6	75.2	134.6	82.6	43.4	91.8	42.6	91.0	41.9	91.0	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.5	21.7	19.9	DB	30.3	108.5	3.3	2.2	27.9	110.1	26.0	111.4	24.2	112.7	21.8	114.3	(4)
			WB	27.0	81.8	2.7	2.1	25.1	83.3	23.6	84.5	22.1	85.7	20.1	87.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.3	58.7	63.3	69.6	77.0	83.3	87.8	88.6	89.0	83.6	76.5	66.8	59.1
	DBStd	13.12	9.18	9.85	8.96	7.07	5.57	3.75	3.12	3.36	4.95	7.45	8.90	9.10
	HDD50	76	30	14	4	0	0	0	0	0	0	1	5	22
	HDD65	768	236	138	60	7	0	0	0	0	0	13	84	230
	CDD50	9320	298	386	611	810	1031	1135	1197	1208	1008	822	510	304
	CDD65	4541	40	89	202	368	567	685	732	743	558	369	140	48
	CDH74	55510	306	721	1767	3757	6839	9470	10354	10574	6650	3698	1053	321
	CDH80	30231	72	238	718	1824	3696	5611	6216	6414	3446	1603	322	71
Wind	WSAvg	10.9	9.1	9.8	11.4	12.3	12.9	13.0	12.8	11.7	9.4	9.8	9.2	8.8
Precipitation	PrecAvg	19.6	1.0	0.8	1.2	1.4	2.4	1.7	2.4	1.6	3.3	1.7	1.1	0.9
	PrecMax	29.0	4.0	2.5	4.3	4.2	5.1	6.2	8.6	8.3	7.0	5.8	3.1	4.2
	PrecMin	11.5	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.6	0.0	0.0	0.1
	PrecStd	4.5	1.1	0.7	1.1	1.1	1.4	1.6	2.2	2.1	1.7	1.6	1.1	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.6	92.2	96.5	102.2	106.0	107.7	106.5	106.5	103.8	97.7	90.6	86.7
		MCWB	65.1	65.4	66.9	71.1	75.3	75.2	74.8	74.7	73.4	75.1	71.5	67.0
	2%	DB	81.5	86.5	91.3	97.3	101.4	104.1	104.0	104.3	99.9	93.9	87.7	81.6
		MCWB	63.3	65.8	67.7	70.4	74.2	75.3	74.5	74.9	74.6	73.9	70.5	65.2
	5%	DB	77.1	82.2	87.9	93.4	98.8	101.6	102.0	102.2	97.2	91.2	83.7	77.5
		MCWB	61.4	64.7	67.6	70.0	74.2	75.5	74.4	75.0	74.5	73.2	68.4	64.0
	10%	DB	72.8	78.5	83.8	90.3	95.1	99.3	99.8	100.1	94.3	88.7	80.7	73.2
		MCWB	60.4	63.4	66.9	69.5	74.0	75.0	74.2	74.8	74.1	72.3	67.1	63.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.0	71.4	73.9	77.1	79.9	80.4	80.4	80.0	80.1	79.2	75.1	71.4
		MCDB	79.1	82.3	85.6	89.0	95.7	94.9	90.4	90.8	90.1	89.1	84.1	80.0
	2%	WB	67.4	69.5	72.2	75.4	78.4	79.1	79.2	78.9	78.5	77.6	73.4	69.7
		MCDB	74.9	79.3	83.1	87.7	92.7	93.6	90.5	91.3	88.4	87.2	81.7	76.0
	5%	WB	65.5	68.2	70.9	74.2	77.4	78.2	78.2	78.2	77.9	76.4	71.7	67.8
		MCDB	72.5	76.8	81.2	86.1	90.8	92.2	90.1	90.8	87.8	85.6	79.7	73.8
	10%	WB	63.4	66.6	69.5	73.0	76.4	77.5	77.5	77.5	77.2	75.3	70.1	65.5
		MCDB	70.2	74.4	78.8	84.3	88.9	90.9	89.7	90.3	86.9	84.1	77.5	72.2
Mean Daily Temperature Range	5% DB	MDBR	20.5	21.0	21.1	22.4	21.7	22.1	22.0	21.8	19.5	19.8	19.5	19.8
		MCDBR	25.7	25.8	25.7	26.1	25.9	25.1	25.2	24.5	22.6	21.2	22.2	24.0
		MCWBR	12.5	10.6	9.5	9.1	6.8	5.1	4.9	4.8	5.5	6.7	9.0	11.7
	5% WB	MCDBR	19.8	20.6	20.9	22.0	22.0	22.3	21.4	21.7	18.8	18.2	18.4	18.7
	MCWBR	11.2	10.5	8.9	8.8	6.6	5.7	5.3	5.3	5.5	7.0	9.0	11.3	
Clear-Sky Solar Irradiance	taub	0.325	0.341	0.364	0.390	0.412	0.404	0.398	0.397	0.396	0.360	0.348	0.325	
	taud	2.517	2.473	2.418	2.365	2.348	2.426	2.445	2.465	2.424	2.517	2.484	2.526	
	Ebn at Noon	293	296	294	288	280	280	282	282	280	286	282	288	
	Edh at Noon	29	33	36	40	40	37	36	35	36	31	30	28	
All-Sky Solar Radiation	RadAvg	1015	1242	1551	1810	1930	2192	2167	2102	1702	1447	1115	921	
	RadStd	126	211	194	176	176	105	161	101	149	167	134	148	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.92	N/A	N/A	+1.55	N/A	+0.63	N/A	N/A	+331	+290
Regional (15 neighbors)		+0.63	N/A	N/A	+0.90	N/A	N/A	N/A	N/A	+241	+211

Nomenclature: See separate page